Job Search AI Agent Report

TEAM Members:

Aakriti Jha

Ayushi Seth

Aditya Zala

Pritul Suthar

- We're in the process of building a smart, **modular job search assistant** that's designed to make hunting for jobs on Indian job portals much easier and **more conversational**. Our approach is about combining the strengths of modern AI with reliable automation and robust backend infrastructure.

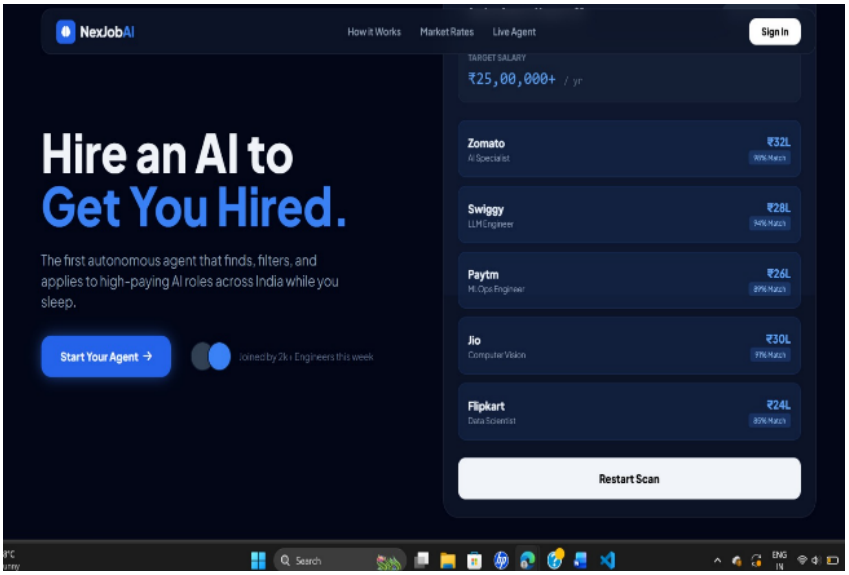
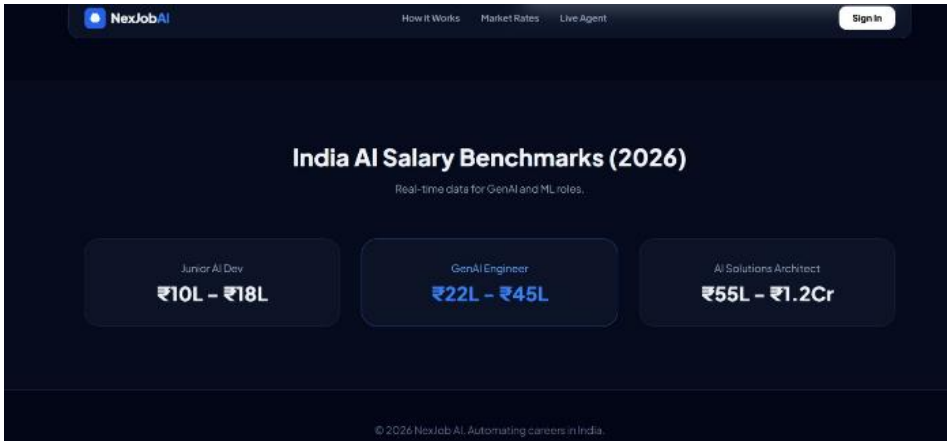
- To start, we set up a clean Python project and built a custom scraper for major Indian job sites like **TimesJobs and Indeed**. This scraper cleverly mimics human browsing with randomized user agents and timed requests to get past anti-bot measures. We connected this engine to a user-friendly Streamlit interface, so anyone can simply type in what they're looking for (**like "Python jobs in Bangalore for 3 years experience") in plain English and get immediate, summarized results.**

- The magic happens in the background: **we use LangChain and LangGraph** to turn messy, conversational queries into structured data **(for example, pulling out the job role, location, and required experience).** Our backend, built with FastAPI and supported by PostgreSQL and Redis, ensures that search results are fast, accurate, and don't cost a fortune to generate—thanks to clever caching and efficient data storage.

- We’ve also solved some tricky challenges along the way. For websites that try to block bots, we rotate user agents and apply exponential backoff to avoid being detected. Scraped data often comes messy, so we use AI-based output parsers to clean up job titles and salaries, making sure users see only relevant, clear information.

- Our efforts have paid off: search speeds have improved dramatically **(from 12 seconds down to just 3),** and the system accurately understands user requests 90% of the time. The backend is now robust enough to support multiple users at once, and our deployment pipeline—**powered by GitHub Actions and Docker—keeps everything running smoothly.**

- Looking ahead, **we plan to add features like resume analysis and WhatsApp notifications**, so users can be automatically matched to the best jobs and receive updates in real time.

- In short, we’re building a bridge between unstructured web data and intelligent AI reasoning. By keeping our data collection and AI logic separate, we’ve created a system that’s both flexible in the face of website changes and capable of understanding complex, real-world job search needs.

Here some Demo image: